

CO4_AT1_DATA INTERPRETATION

1. Real-Time Detection System

A surveillance system uses the following techniques:

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Compare the three techniques and evaluate which method should be selected for a surveillance system operating under strict latency constraints while maintaining acceptable accuracy. Justify your decision considering deployment conditions.

2. Robust Object Detection

Method	Normal	Occlusion	Low Light
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Compare the robustness of the three methods under normal, occluded, and low-light conditions. Evaluate which method provides the best overall performance and discuss the trade-offs involved.

3. Optical Flow Analysis

Two optical flow methods are available:

- Method A: Produces stable motion vectors but misses small motions.
- Method B: Detects fine motion but produces noisy vectors.

Compare the two methods and evaluate which method is more suitable for motion tracking in dynamic scenes. Justify your answer based on accuracy, stability, noise sensitivity, and application requirements.

4. Model Generalization and Overfitting – 10 Marks

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Compare the two models and evaluate which model is more reliable for real-world deployment. Explain the generalization ability and overfitting risk of each model and justify your selection.

5. System Performance under Constraints

Model	Accuracy	Latency	Cost
A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

Compare the three models and evaluate which model provides the best balance between accuracy, latency, and cost for a real-time application. Justify your choice considering practical deployment constraints.

1. Real-Time Detection System

The three methods are compared based on accuracy and speed:

- **Viola-Jones:** 78% accuracy and fast.
- **YOLO:** 92% accuracy and moderate speed.
- **Deep Learning:** 95% accuracy but slow.

For a surveillance system with **strict latency constraints**, **Viola-Jones** is the most suitable when very fast response is the main requirement. However, its accuracy is lower than YOLO and Deep Learning.

YOLO is the better practical choice if the system requires a balance between speed and accuracy. It provides **92% accuracy** with moderate speed, making it suitable for real-time surveillance.

Conclusion: YOLO should be selected because it offers significantly better accuracy than Viola-Jones while still maintaining reasonable speed. Deep Learning has the highest accuracy but is less suitable when strict latency is required.

2. Robust Object Detection

Method	Normal Occlusion Low Light		
Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Deep Learning provides the best overall performance under all three conditions.

- Under normal conditions, Deep Learning achieves **95%**.
- Under occlusion, it achieves **85%**.
- Under low-light conditions, it achieves **80%**.

YOLO performs second best, while Viola-Jones has the lowest robustness.

Conclusion: Deep Learning is the best choice for robust object detection because it maintains high accuracy even with occlusion and low-light conditions. The main trade-off is that it generally requires greater computational resources and may have higher latency than simpler methods.

3. Optical Flow Analysis

Method A produces stable motion vectors but may miss small movements. Therefore, it has better stability and lower noise sensitivity.

Method B detects fine movements more effectively, but its vectors are noisy. Therefore, it provides better sensitivity to small motion but may reduce tracking stability.

For **dynamic scenes**, **Method A** is more suitable when stable and reliable tracking is the priority. Its stable motion vectors make it less sensitive to noise.

However, if the application requires detection of **very small or fine movements**, Method B may be preferred despite its noise.

Conclusion: Method A is the better general-purpose choice for dynamic-scene motion tracking because stability and reliable motion estimation are important. Method B is preferable when detecting small movements is more important than noise-free tracking.

4. Model Generalization and Overfitting

Model Training Accuracy Testing Accuracy

A	98%	70%
B	90%	88%

Model A has very high training accuracy of **98%**, but its testing accuracy drops to **70%**. This large difference indicates **overfitting**. Model A has learned the training data very well but does not generalize effectively to unseen data.

Model B has **90% training accuracy** and **88% testing accuracy**. The small difference indicates good generalization and low overfitting risk.

Conclusion: **Model B is more reliable for real-world deployment** because its training and testing performances are close. It generalizes better to unseen data and has a much lower risk of overfitting than Model A.

5. System Performance under Constraints

Model Accuracy Latency Cost

A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

For a real-time application:

- **Model A:** Lowest accuracy, but fastest latency and lowest cost.
- **Model B:** Good accuracy, moderate latency, and moderate cost.
- **Model C:** Highest accuracy but highest latency and cost.

Model B provides the best overall balance. It achieves **92% accuracy**, which is considerably better than Model A, while its **80 ms latency** is still lower than Model C's 120 ms. Its moderate cost is also more practical than Model C's high cost.

Conclusion: Model B should be selected for a real-time application because it provides the best compromise between accuracy, latency, and cost. Model A is preferable when cost and speed are the highest priorities, while Model C is suitable when maximum accuracy is more important than latency and cost.